

TEST CASES LIST

CONVERTER ENGINE/MODULE TEST CASES

Normal Cases:

#	Input	Sign	Exponent	Coefficient	Hex
1	123.456	0	1000 0101	111 0110 110 1001 0111 1000	42F6 E978
2	-123.456	1	1000 0101	111 0110 110 1001 0111 1000	C2F6 E978
3	0.5	0	0111 1110	000 0000 0000 0000 0000 0000	3F00 0000
4	1	0	0111 1111	000 0000 0000 0000 0000 0000	3F80 0000
5	-7.5	1	1000 0001	111 0000 0000 0000 0000 0000	C0F0 0000

Special Cases:

#	Input	Expected Special Case	Sign	Exponent	Coefficient	Hex	Decoded
1	0	+0	0	0000 0000	000 0000 0000 0000 0000 0000	0000 0000	0
2	-0	-1	1	0000 0000	000 0000 0000 0000 0000 0000	8000 0000	0

3	NaN	NaN	0	1111 1111	100 0000 0000 0000 0000 0000	7FC0 0000	NaN
4	Infinity	Infinity	0	1111 1111	000 0000 0000 0000 0000 0000	7F80 0000	Infinity
5	-Infinity	-Infinity	1	1111 1111	000 0000 0000 0000 0000 0000	FF80 0000	-Infinity
6	1e40	Overflow	0	1111 1111	000 0000 0000 0000 0000 0000	7F80 0000	Infinity
7	-1e40	Overflow	1	1111 1111	000 0000 0000 0000 0000 0000	FF80 0000	-Infinity
8	1e-40	Underflow	0	0000 0000	000 0000 0000 0000 0000 0000	0000 0000	0
9	-1e-40	Underflow	1	0000 0000	000 0000 0000 0000 0000 0000	8000 0000	-1

Input Validation:

#	Input	What should appear
1	(empty box)	"Please enter a decimal value."
2	abc	"abc" is not a valid decimal number.
3	1234abc	"1234abc" is not a valid decimal number.
4	-1234abc	"-1234abc" is not a valid decimal number.

ROUNDING MODULE TEST CASES

Normal Cases:

#	Input	Input Type	Target Digits/Bits	Purpose	Expected Output
1	123.456789	Decimal	5	Normal positive decimal rounding	C: 123.45 U: 123.46 D: 123.45 N: 123.46
2	-123.456789	Decimal	5	Negative decimal rounding	C: -123.45 U: -123.45 D: -123.46 N: -123.46
3	101.1011	Binary	5	Normal positive binary rounding	C: 101.10 U: 101.11 D: 101.10 N: 101.11
4	-101.1011	Binary	5	Negative binary rounding	C: -101.10 U: -101.10 D: -101.11 N: -101.11
5	3.14159265	Decimal	4	Fractional precision rounding	C: 3.141 U: 3.142 D: 3.141 N: 3.142

Boundary Cases:

#	Input	Input Type	Target Digits/Bits	Purpose	Expected Output
1	0	Decimal	5	Zero Handling	All: 0
2	0.0000012345	Decimal	4	Very small number handling	C: 0.000001234 U: 0.000001235 D: 0.000001234 N: 0.000001234
3	999.999	Decimal	3	Carry after rounding	C: 999 U: 1000 D: 999 N: 1000
4	111.111	Binary	3	Binary carry after rounding	C: 111 U: 1000 D: 111 N: 1000
5	1.234560	Decimal	6	No rounding required	All: 1.234560

Special Rounding Cases:

#	Input	Input Type	Target Digits/Bits	Purpose	Expected Output
1	12.345	Decimal	4	Tie-to-even where retained digit is even	C: 12.34 U: 12.35 D: 12.34 N: 12.34
2	12.355	Decimal	4	Tie-to-even where retained digit is odd	C: 12.35 U: 12.36 D: 12.35 N: 12.36
3	101.101	Binary	5	Binary tie-to-even	C: 101.10 U: 101.11 D: 101.10 N: 101.10
4	101.111	Binary	5	Binary tie causing carry	C: 101.11 U: 110.00 D: 101.11 N: 110.00
5	-12.355	Decimal	4	Negative tie-to-even	C: -12.35 U: -12.35 D: -12.36 N: -12.36

Input Validation:

#	Input	Input Type	Target Digits/Bits	Expected Output
1	(empty)	Decimal	any	"Input cannot be empty."
2	abc	Decimal	any	"Invalid decimal number."
3	10201	Binary	any	"Invalid binary number."
4	12.34	Decimal	0	"Digits must be greater than 0."
5	101.01	Binary	-1	"Digits must be greater than 0."

ARITHMETIC ENGINE/MODULE TEST CASES: DIVISION

INPUT A	INPUT B	FINAL DECIMAL	FINAL BINARY	FINAL HEX
10	3	3.3333332538 604736	0100 0000 0101 0101 0101 0101 0101 0101	40555555
-7.5	2	-3.75	1100 0000 0111 0000 0000 0000 0000 0000	C0700000
1	10	0.1000000014 9011612	0011 1101 1100 1100 1100 1100 1100 1101	3DCCCCCD
0	0	NaN	0111 1111 1100 0000 0000 0000 0000 0000	7FC00000
5	0	Infinity	0111 1111 1000 0000 0000 0000 0000 0000	7F800000

ARITHMETIC ENGINE/MODULE TEST CASES: SUBTRACTION

INPUT A	INPUT B	FINAL DECIMAL	FINAL BINARY	FINAL HEX
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6	7	-1.0	1011 1111 1000 0000 0000 0000 0000 0000	BF800000
10	3	7.0	0100 0000 1110 0000 0000 0000 0000 0000	40E00000
-7.5	2	-9.5	1100 0001 0001 1000 0000 0000 0000 0000	C1180000
10	Infinity	-Infinity	1111 1111 1000 0000 0000 0000 0000 0000	FF800000
0	0	+0	0000 0000 0000 0000 0000 0000 0000 0000	00000000